

strongest predictive power is principal component analysis. These matrices are as follows:

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} \quad V = \begin{bmatrix} v_{11} & v_{21} & \cdots & v_{n1} \\ v_{12} & v_{22} & \cdots & v_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ v_{1n} & v_{2n} & \cdots & v_{nn} \end{bmatrix}$$

Since D is a diagonal matrix, we have $D = D^{1/2}D^{1/2}$, $D = D'$, and $D^{1/2} = (D^{1/2})'$

Then, our covariance matrix C can be written as:

$$\bar{C} = VDV' = VD^{1/2}D^{1/2}V' = VD^{1/2}(VD^{1/2})'$$

Step 3. Compute β in terms of the eigenvalues and eigenvectors:

$$\beta = (VD^{1/2})'$$

Then the full sample covariance matrix expressed in terms of β is:

$$\beta' \beta = VD^{1/2}(VD^{1/2})' \quad (6.43)$$

Step 4. Remove spurious relationship due to data limitation.

To remove the potential spurious relationship we only use the eigenvalues and eigenvectors with the strongest predictive power.

How many factors should be selected?

In our eigenvalue-eigenvector decomposition each eigenvalue λ_k of the sample covariance matrix explains exactly $\lambda_k / \sum \lambda$ percent of the total variance. Since the eigenvalues are sorted from highest to lowest, a plot of the percentage of variance explained will show how quickly the predictive power of the factors declines. If the covariance matrix is generated by say 10 factors then the first 10 eigenvalues should explain the large majority of the total variance.

There are many way to determine how many factors should be selected to model returns. For example, some analysts will select the minimum number of factors that explain a pre-specified amount of variance, some will select the number of factors up to where there is a break-point or fall-off in explanatory power. And others may select factors so that the variance $> 1/n$. Assuming that each factor should explain at least $1/n$ of the total. Readers can refer to Dowd (1998) for further techniques.